

MATH 330 Homework 4

Relations

Exercise 1. Determine whether or not each of the following relations is reflexive, symmetric, antisymmetric, and/or transitive. Are any of these equivalence relations?

1. The relation $\leq$ on $\mathbb{R}$.
2. Equality on $\mathbb{R}$, i.e. the set of ordered pairs of real numbers whose first and second coordinates are equal.
3. The relation $|$ on $\mathbb{N}$, where $a | b$ means that $b = an$ for some $n \in \mathbb{N}$.
4. The relation $\sim$ on $\mathbb{N}$ defined by $a \sim b$ if there is an integer $n > 1$ that evenly divides both a and b .

Exercise 2. Define the relation $\equiv_5$ on $\mathbb{Z}$ by: $a \equiv_5 b$ if there is an integer n so that $b - a = 5n$. Show that $\equiv_5$ is an equivalence relation on $\mathbb{Z}$. Note: this is a fairly common equivalence relation. The phrase $a \equiv_5 b$ is usually read “ a is equivalent to b modulo 5.”

Exercise 3. A relation on a set X is said to be a partial order on X if it is reflexive, antisymmetric, and transitive. Let $X = P(\mathbb{R})$ and consider the inclusion relation defined by $A \subseteq B$.

1. Show that $\subseteq$ is a partial order on X .
2. Is strict inclusion $\subsetneq$ a partial order on X ?

Exercise 4. Define a relation $\sim$ on $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ by $(x, y) \sim (w, z)$ if $x^2 + y^2 = w^2 + z^2$. Show that $\sim$ is an equivalence relation on $\mathbb{R}^2$.

Exercise 5. Consider the equivalence relation defined in Exercise 4. For each point $(x, y) \in \mathbb{R}^2$, the equivalence class $[(x, y)]$ is a familiar geometric figure in $\mathbb{R}^2$. What is it?